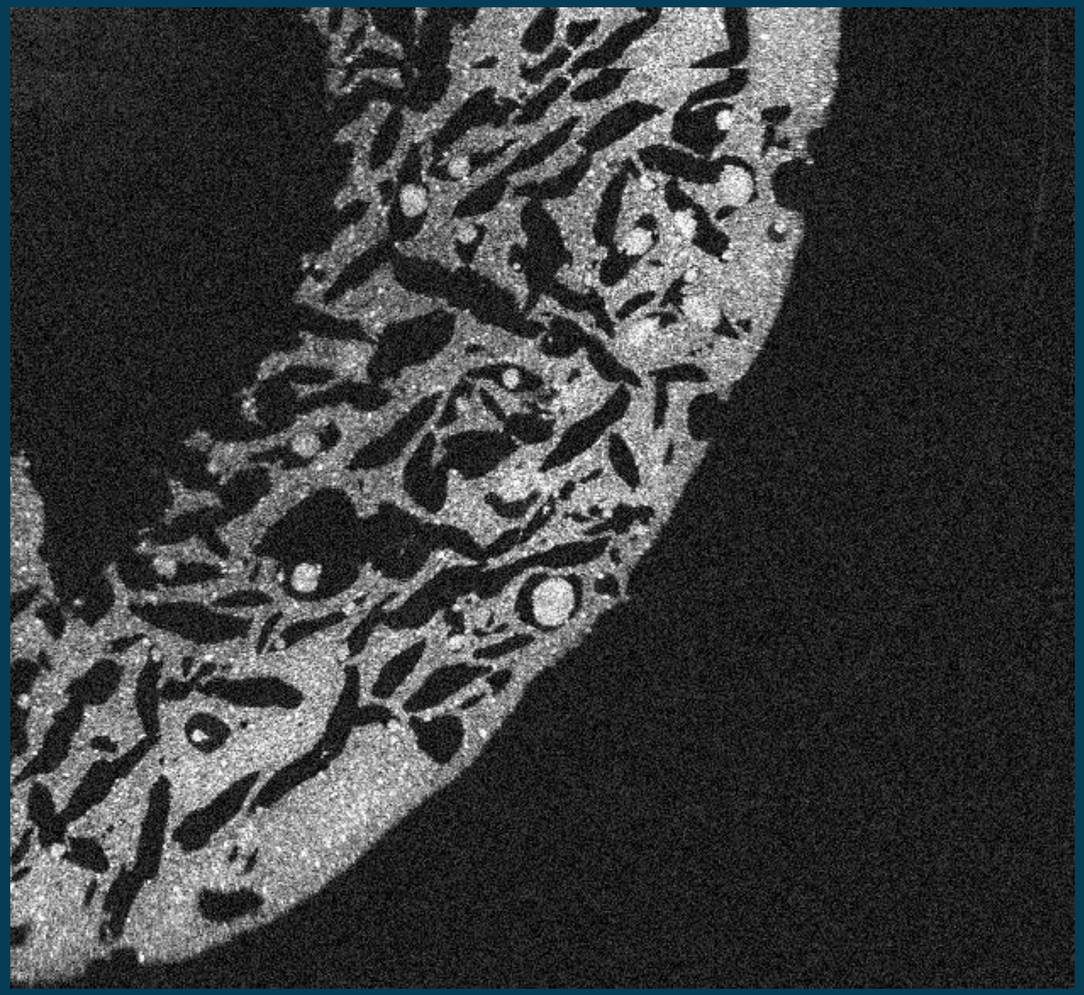
Final Project



Neil SI SMAIL

SUMMARY

01 Context and Challenges

02 Image Processing

03 Metric computation

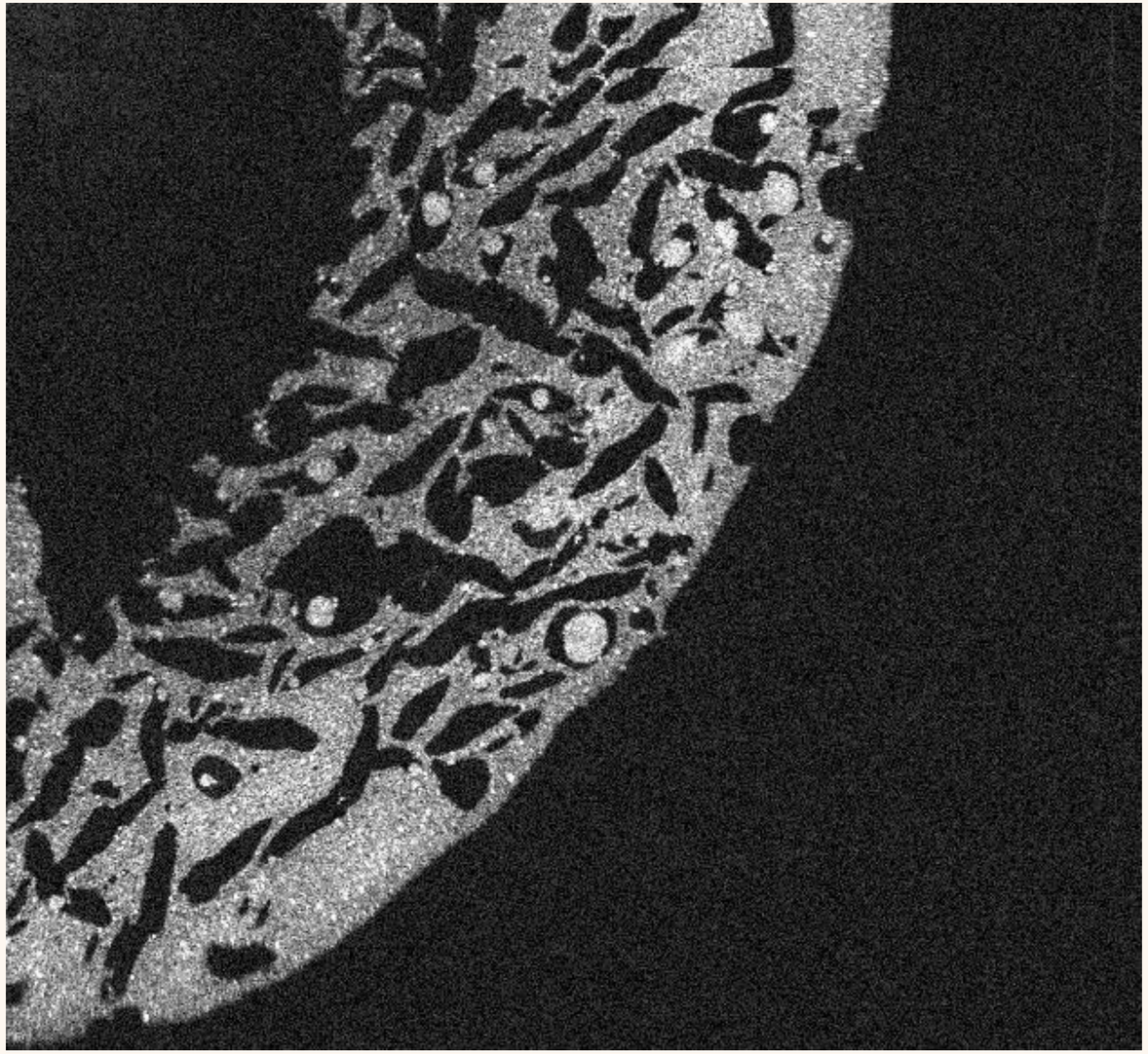
04 Conclusion - Future Work

CONTEXT

Stem cell culture

Black and white Tomography

Focusing of a 2D sample



3. Effective Solutions

A.

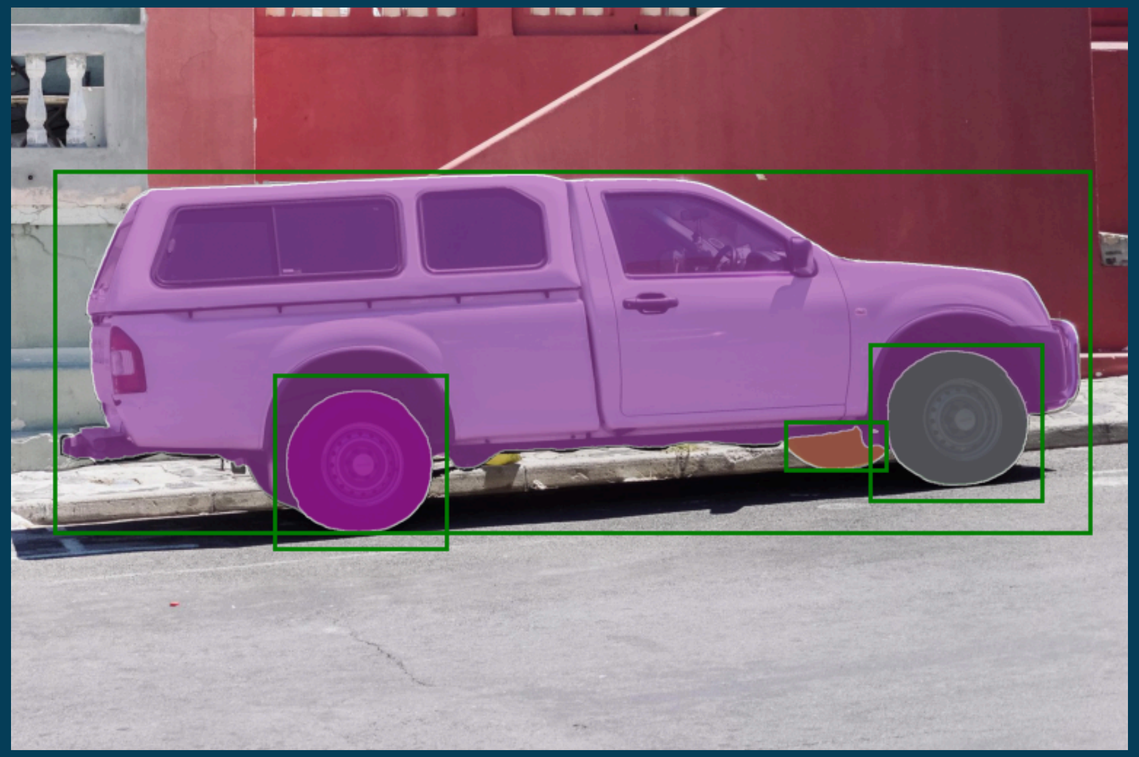
Use of Machine Learning

Fine-tuning of MedSAM

B.

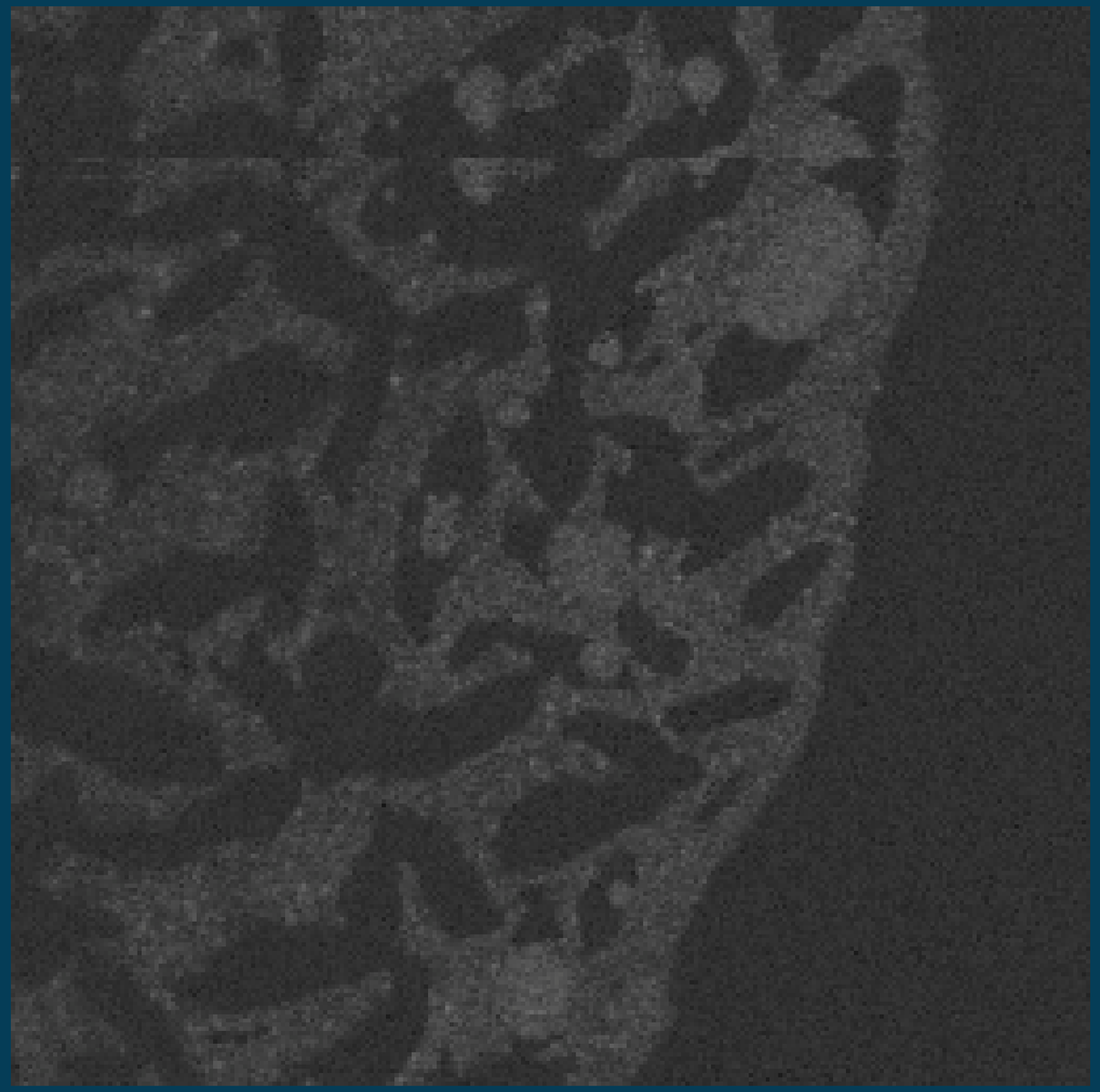
Use of Image processing

SAM2



Use case

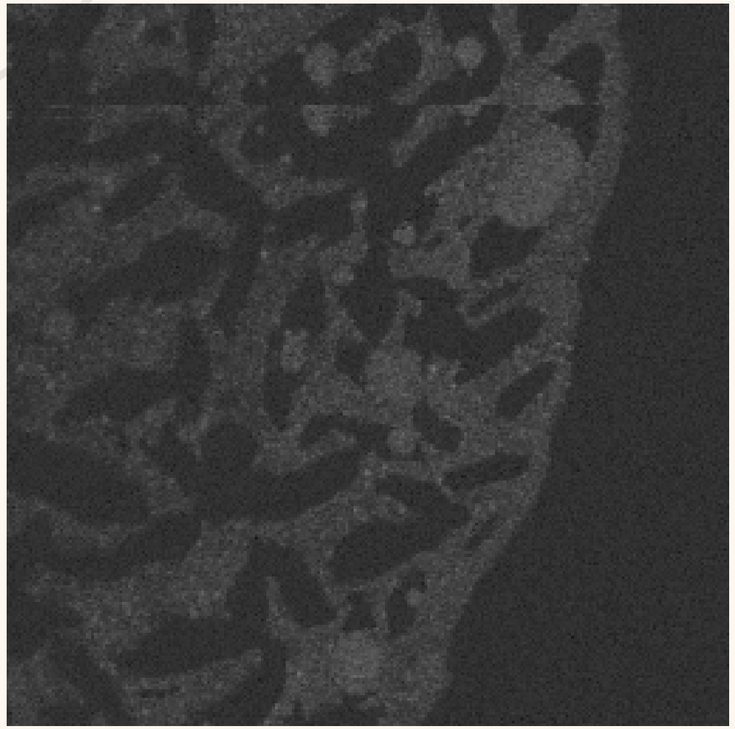
Medical segmentation : organoids seeking



Challenges :

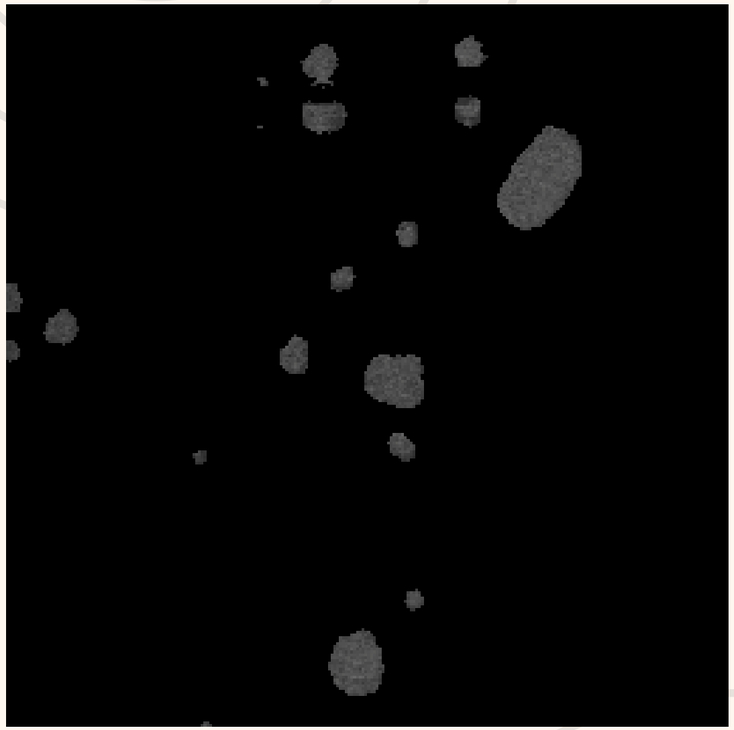
- Poor contrast between the cells and the environment
- Very noisy image
- Random sized organoids

PREPROCESSING



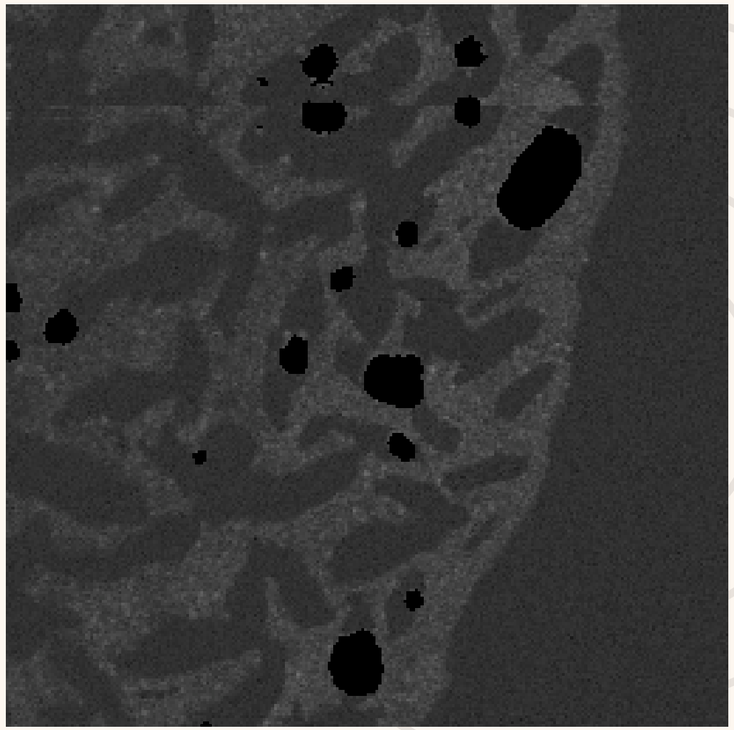
Original image

Binary Mask

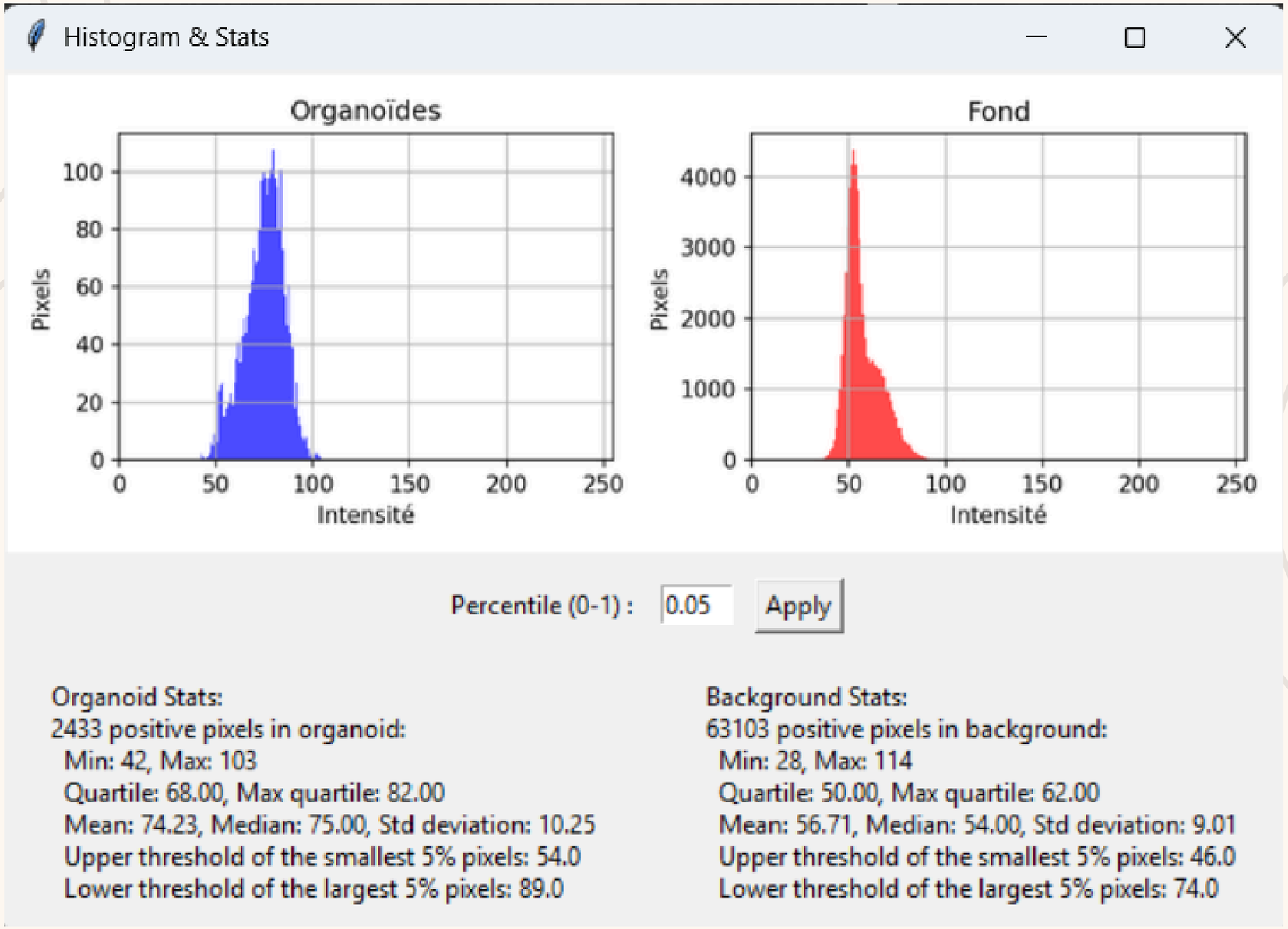


organoids only

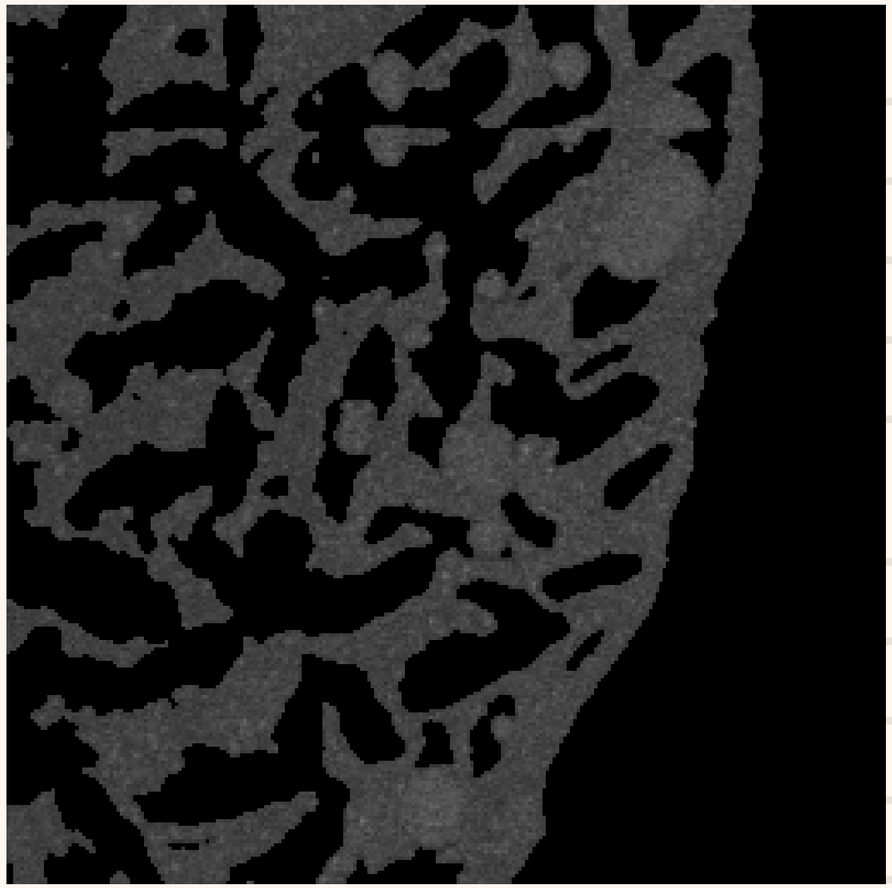
inverted mask



background only


Histogram & Stats
Organoïdes
Pixels
Intensité
Fond
Pixels
Intensité
Percentile (0-1) :
0.05
Apply
Organoid Stats:
2433 positive pixels in organoid:
Min: 42, Max: 103
Quartile: 68.00, Max quartile: 82.00
Mean: 74.23, Median: 75.00, Std deviation: 10.25
Upper threshold of the smallest 5% pixels: 54.0
Lower threshold of the largest 5% pixels: 89.0
Background Stats:
63103 positive pixels in background:
Min: 28, Max: 114
Quartile: 50.00, Max quartile: 62.00
Mean: 56.71, Median: 54.00, Std deviation: 9.01
Upper threshold of the smallest 5% pixels: 46.0
Lower threshold of the largest 5% pixels: 74.0

Unwanted background



cleaned image

Quick review of Yen's Threshold method

Yen's method uses the **histogram of image intensities** and computes a threshold `T` that **maximizes** the following criterion:

$$T = \arg\max_t \left(-\log(P_1(t)^2 + P_2(t)^2) + 2 \cdot \log(P_1(t) \cdot P_2(t)) \right)$$

Where:

- $P_1(t)$ = cumulative probability of intensity ≤ t
- $P_2(t)$ = cumulative probability of intensity > t

It essentially finds the threshold that **maximizes inter-class entropy**, favoring strong contrast between two distributions.

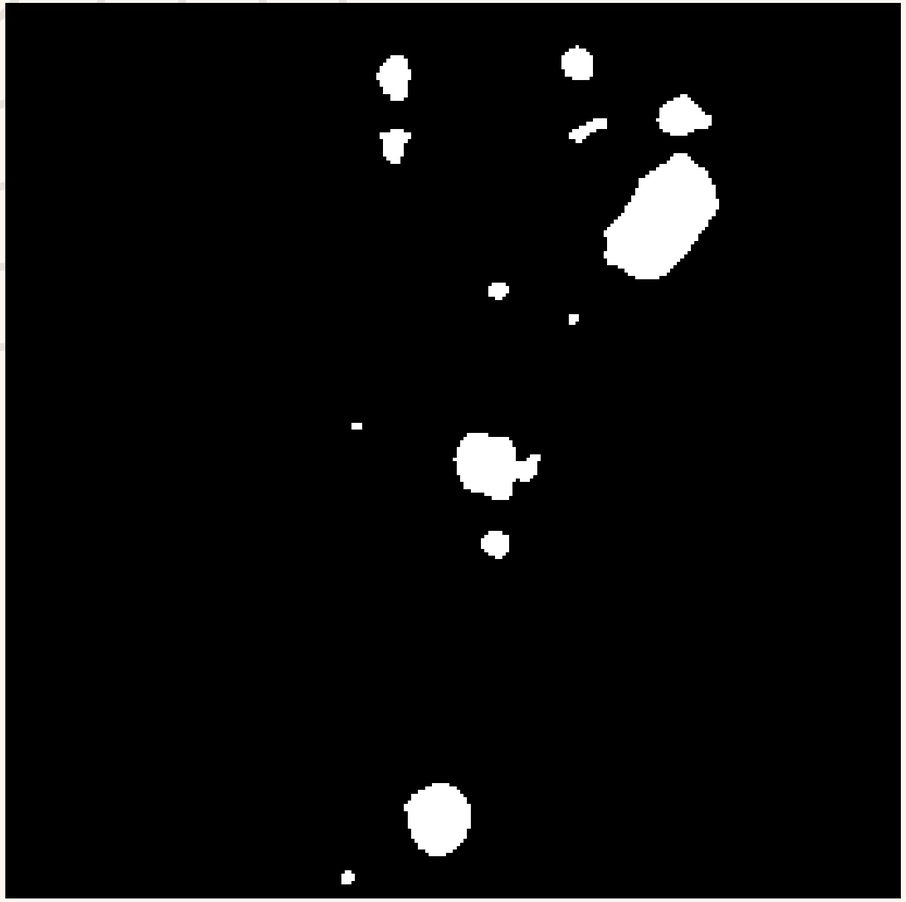


Image processing prediction

Truth : human made binary mask

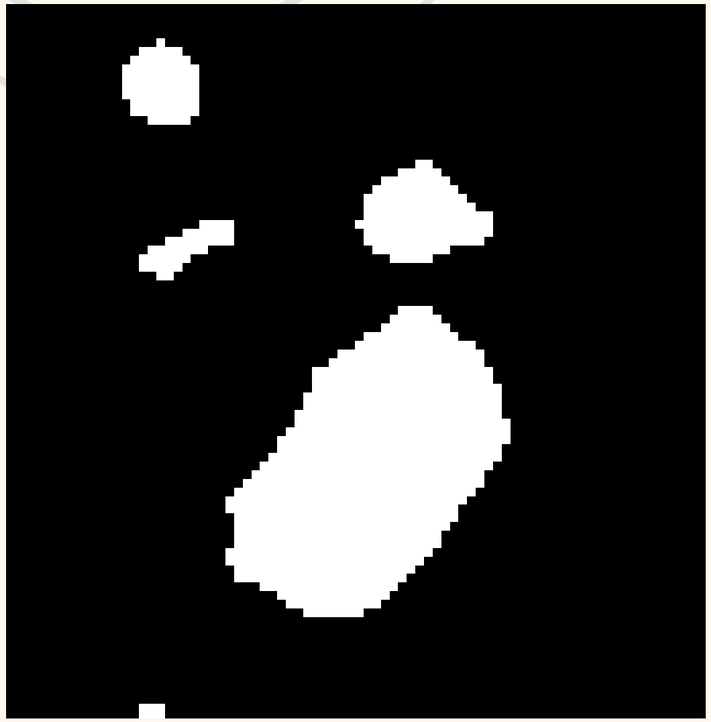
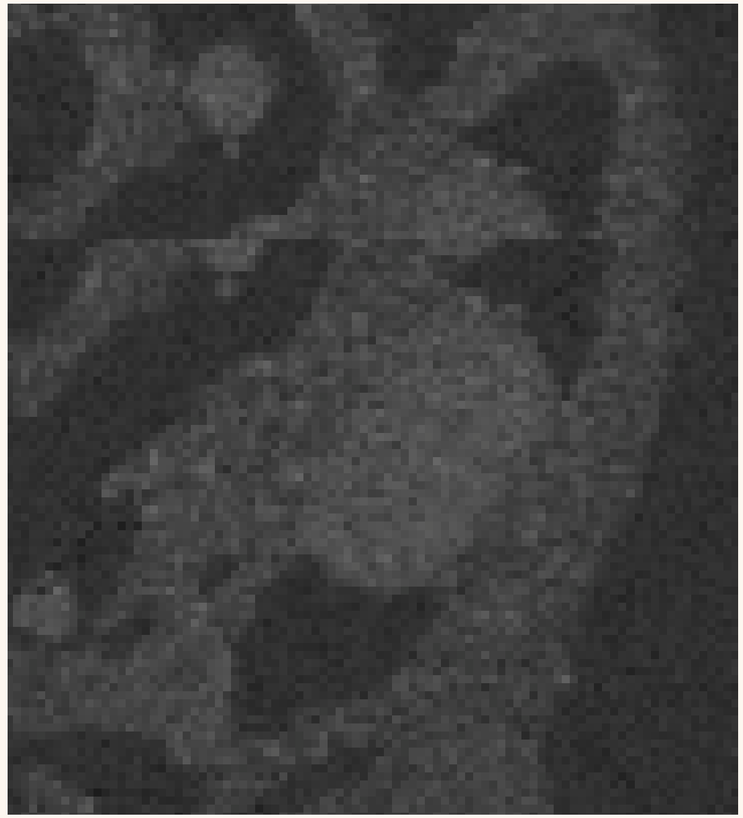
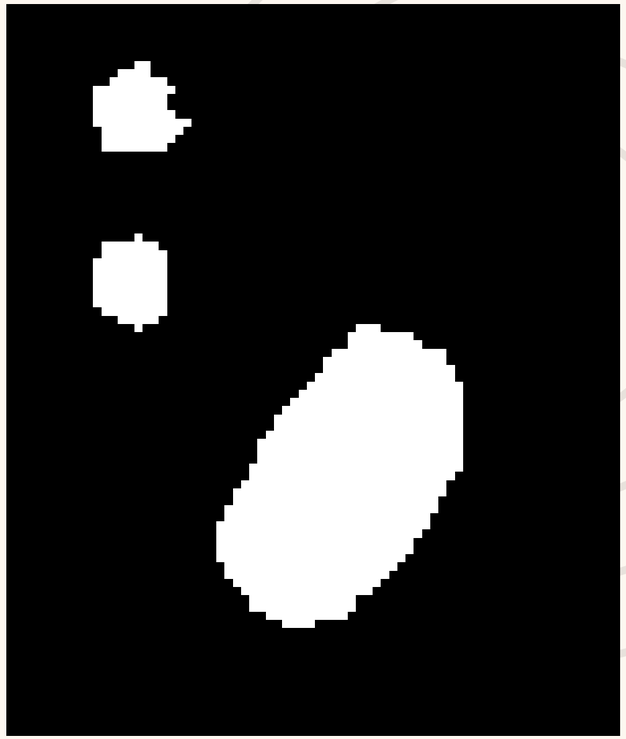


Image processing prediction

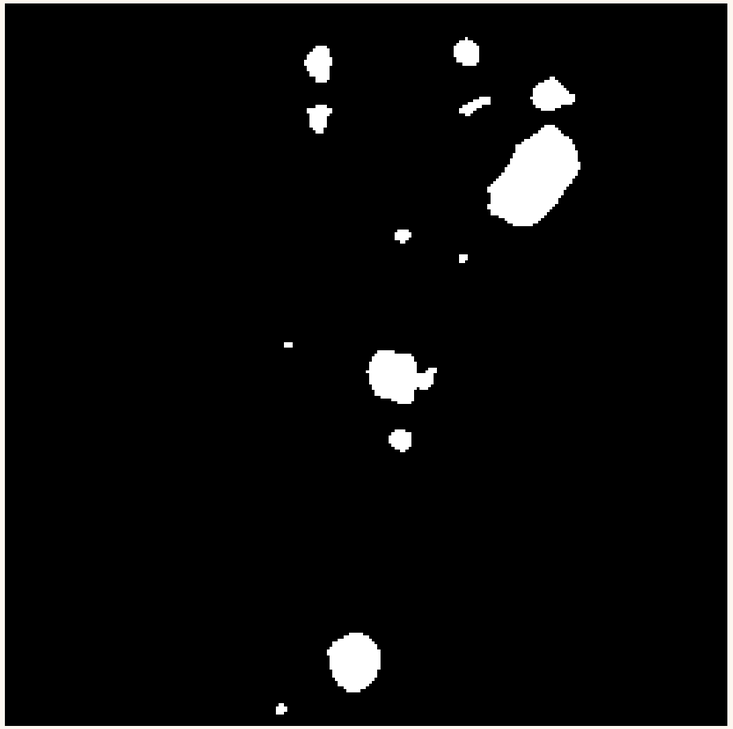


Human mistake



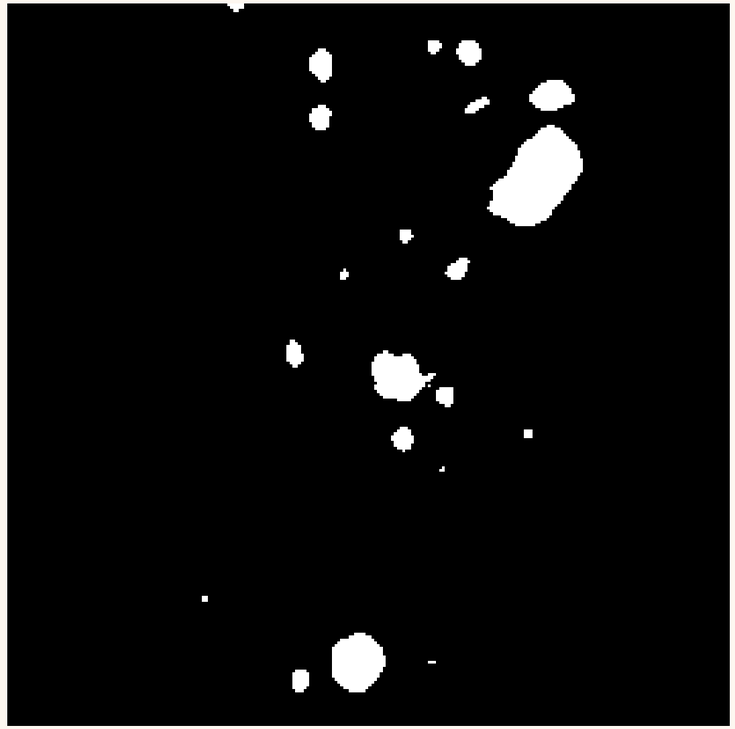
Truth : human made binary mask

The method effectivness on an uncleaned image



Prediction on an cleaned image

Truth mask



Prediction on an uncleaned image

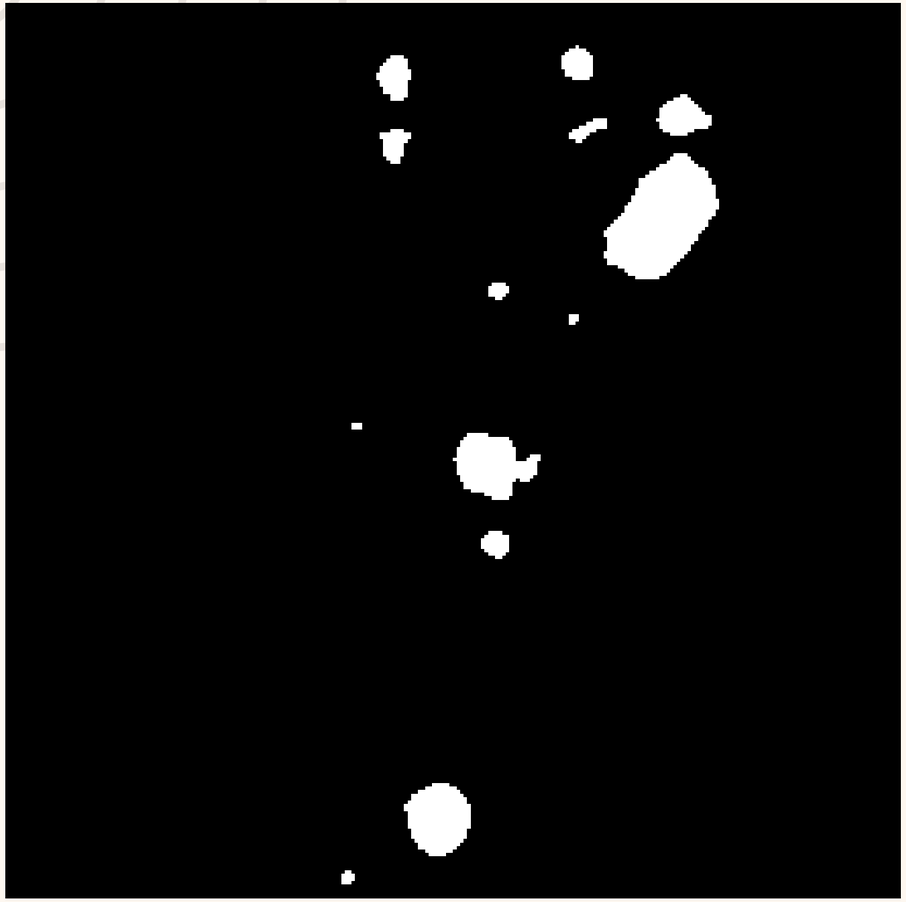
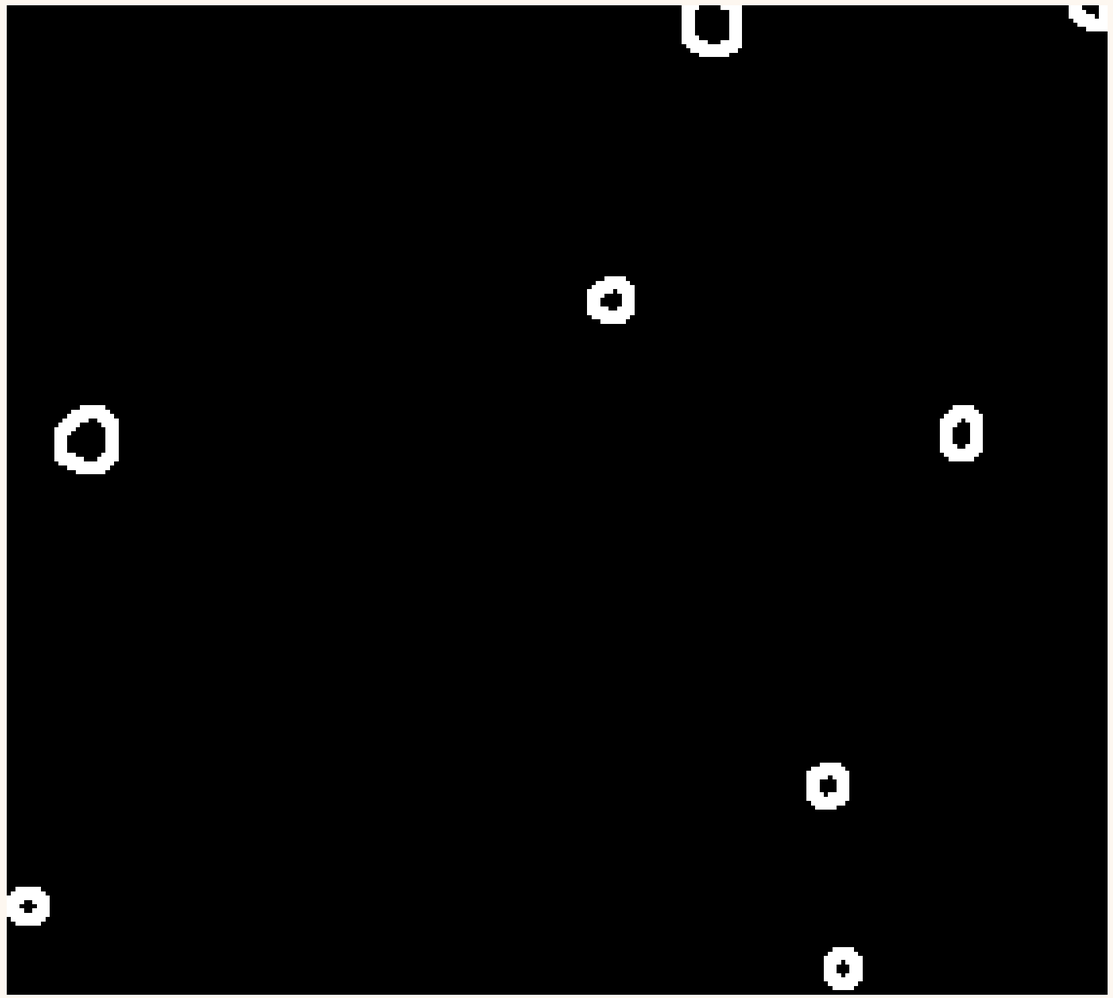


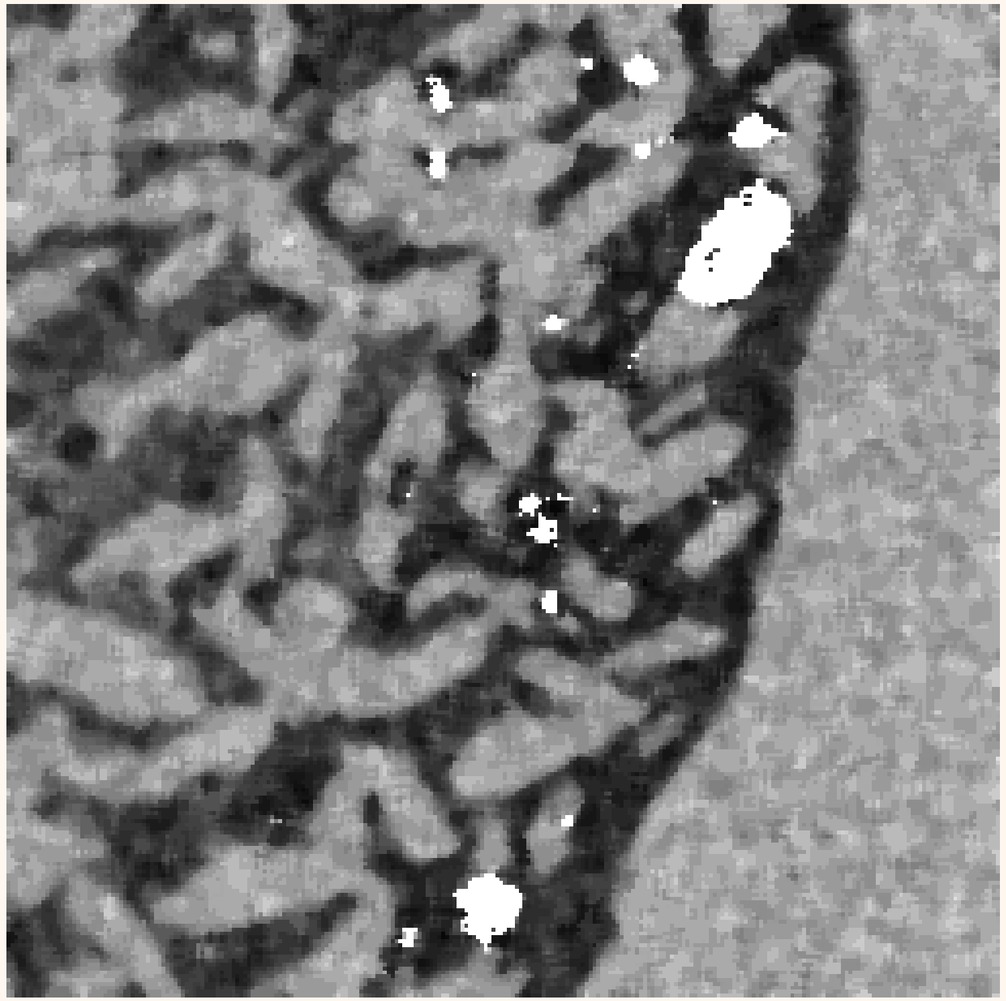
Image processing prediction

Truth : human made binary mask

The Elliptic test



The Magical combo : Z-normalisation/Gamma correction



Z-score normalisation

Z-score normalization rescales a dataset so that it has:

- **Mean (μ) = 0**
- **Standard Deviation (σ) = 1**

🧠 **Formula:**

$$z = \frac{x - \mu}{\sigma}$$

- x: Original value
- μ: Mean of the dataset
- σ: Standard deviation of the dataset
- z: Normalized value (Z-score)

Gamma Correction

🧮 **Formula:**

$$I_{out} = I_{in}^{\gamma}$$

- I_{in}: Normalized input pixel value (0–1)
- γ: Gamma value
- I_{out}: Output pixel value after correction

📊 **Interpretation:**

- $\gamma < 1$ → Lightens the image
- $\gamma > 1$ → Darkens the image

A METRIC : F1 SCORE

- **Precision**

  = proportion of predicted positives that are **correct**

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall**

  = proportion of actual positives that are **retrieved**

$$\text{Recall} = \frac{TP}{TP + FN}$$

(where TP = True Positives, FP = False Positives, FN = False Negatives)

F1 Score: Harmonic Mean

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Penalizes imbalance between precision and recall

Always between **0 (worst)** and **1 (best)**

computing f1-score with intensity method - comparison with MedSAM

undesired	organoids	base
100	1539	2433
FP	TP	FN
100	1539	894

precision	recall	f1
0,93898719	0,6325524	0,75589391

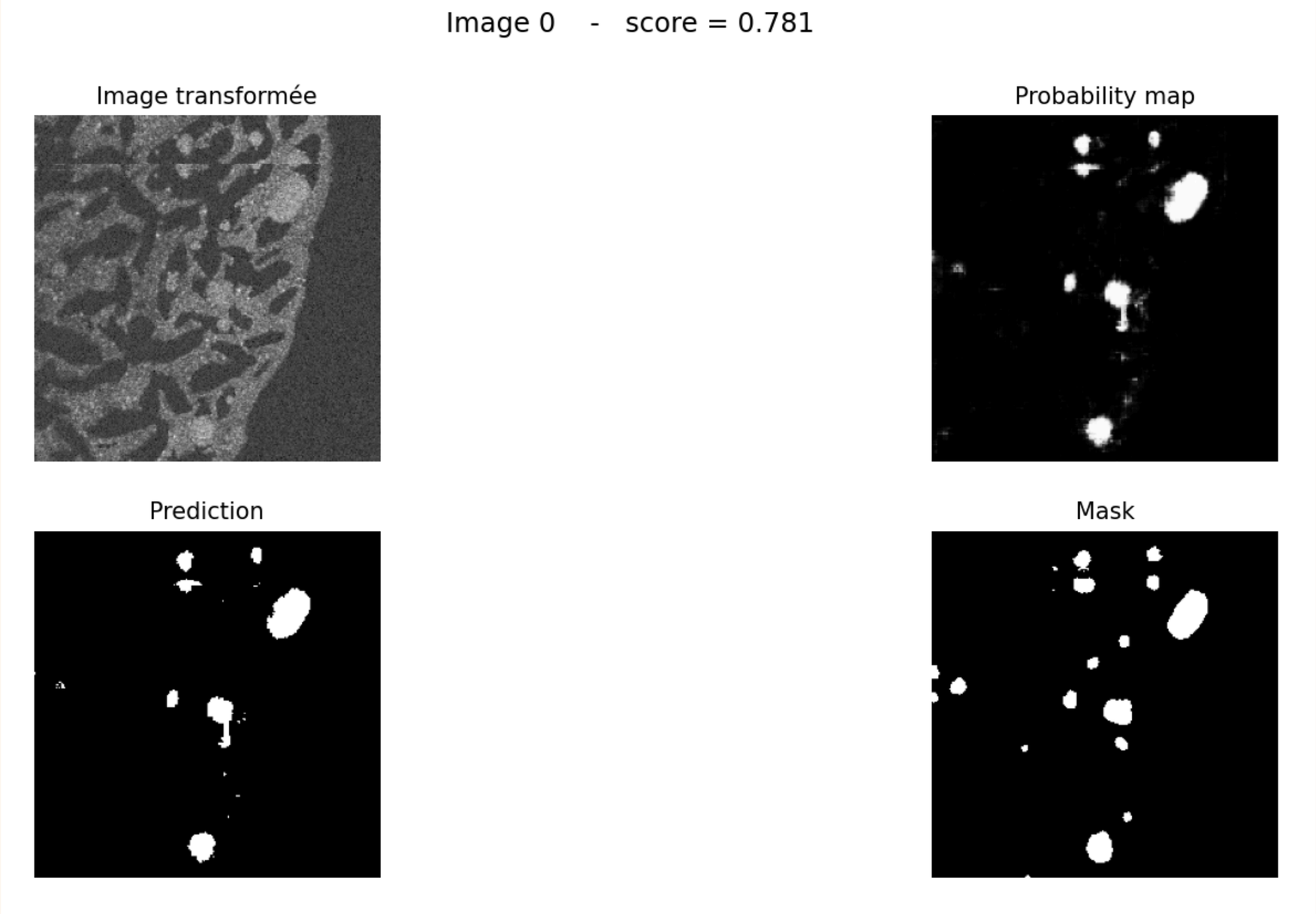


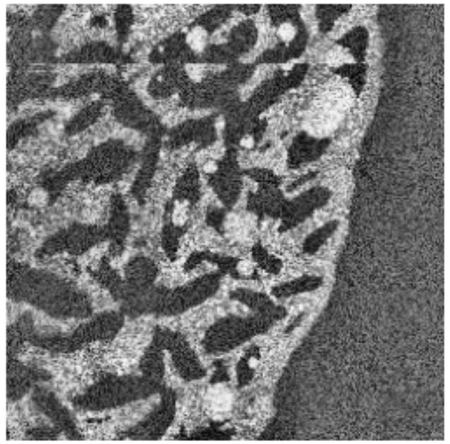
Image 0 - score = 0.781
Image transformée
Probability map
Prediction
Mask

Evaluation f1 score with MedSAM on different algorithms

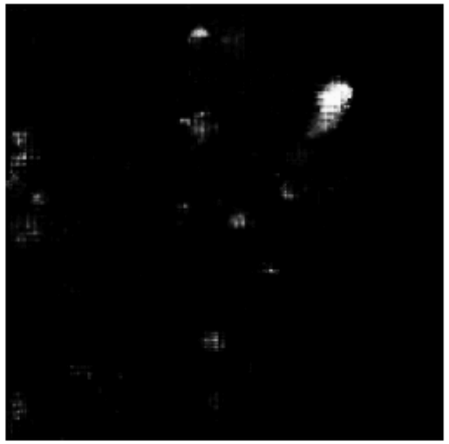
Algorithm	score
raw	0.7832071576049553
gaussian_blur	0.612684883429431
median_blur	0.5921579087756735
bilateral_filter	0.5137759118341643
clahe	0.4280442804428044
histogram_equalization	0.2413913043478260 7
sobel_edge	0.0128824476650563 6
canny	0.0402090872537193 4
laplacian	0.1482269503546099 2
unsharp_mask	0.749766136576239 5
adaptive_threshold	0.1343228200371057 4

29 : equalize_adapthist - score = 0.228

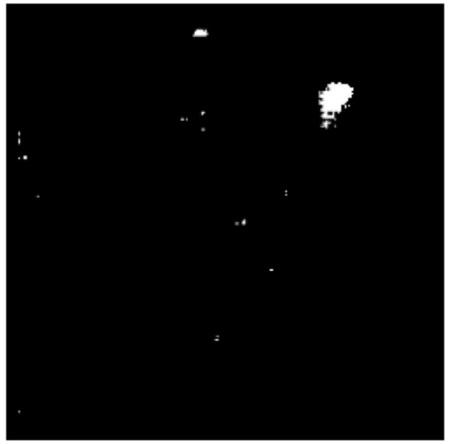
Image transformée



Probability map



Prediction



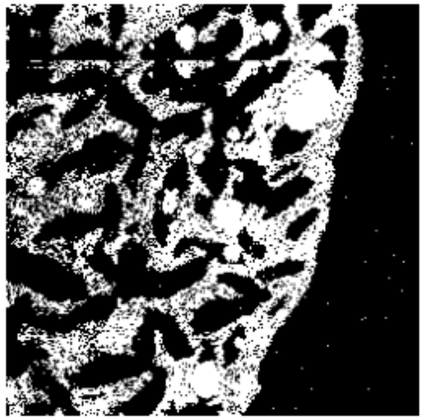
Mask

Précédent

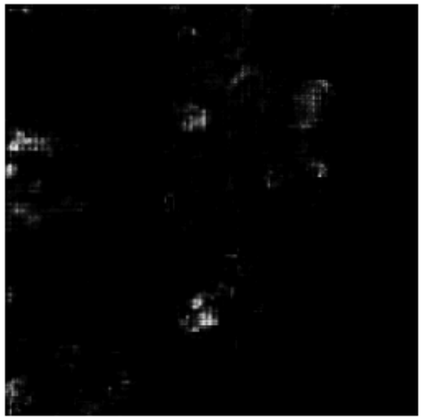
Suivant

11 : otsu_threshold - score = 0.016

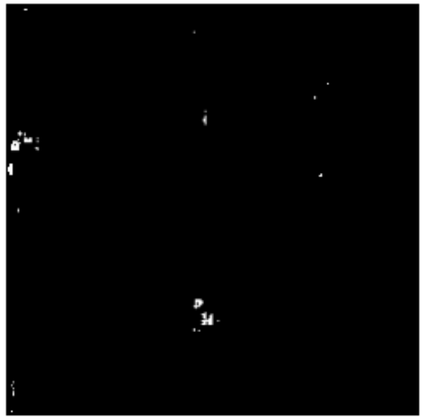
Image transformée



Probability map



Prediction



Mask

Précédent

Suivant

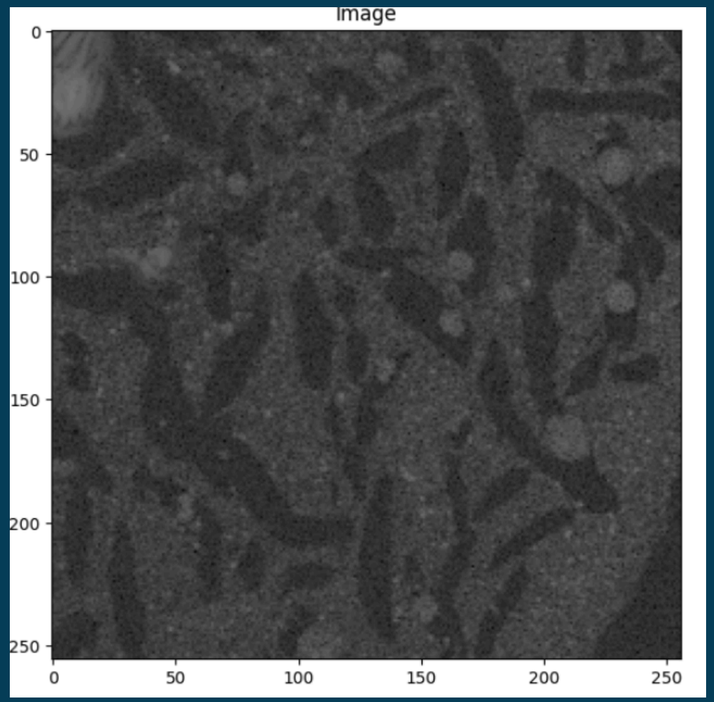
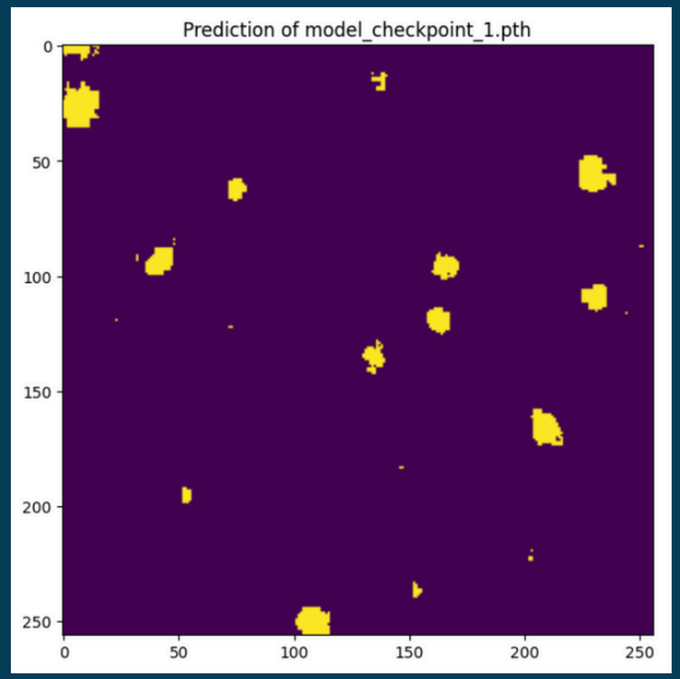
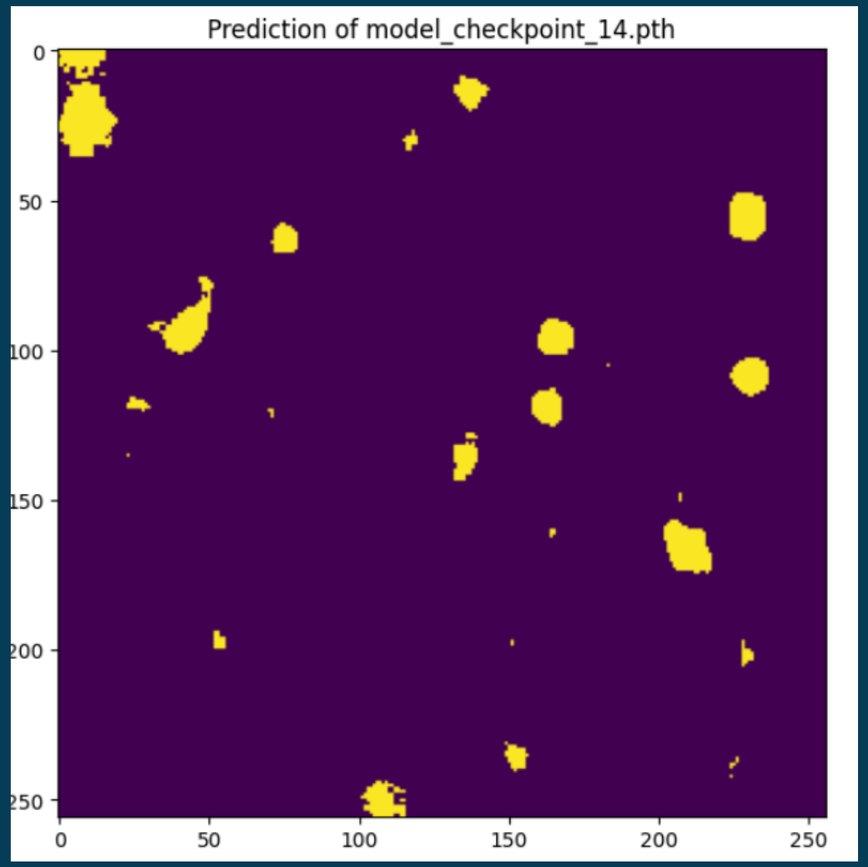
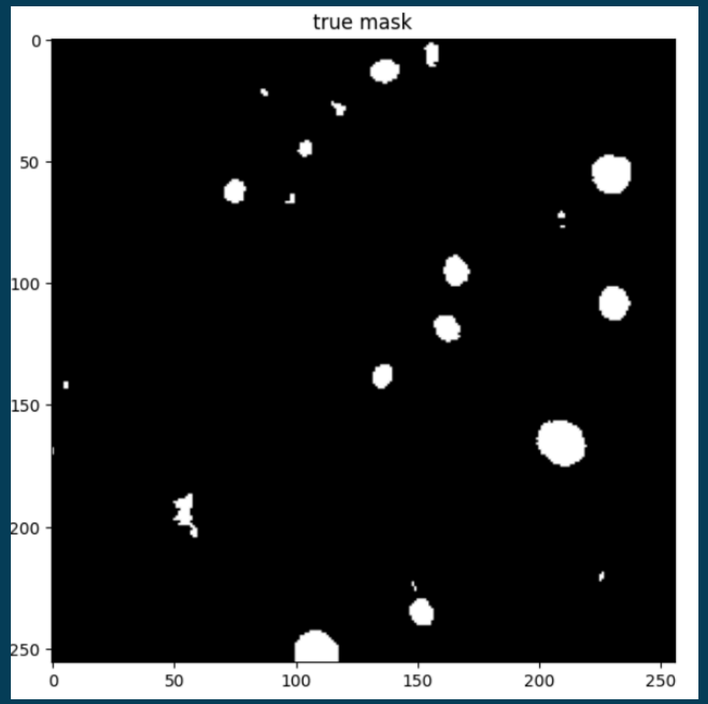
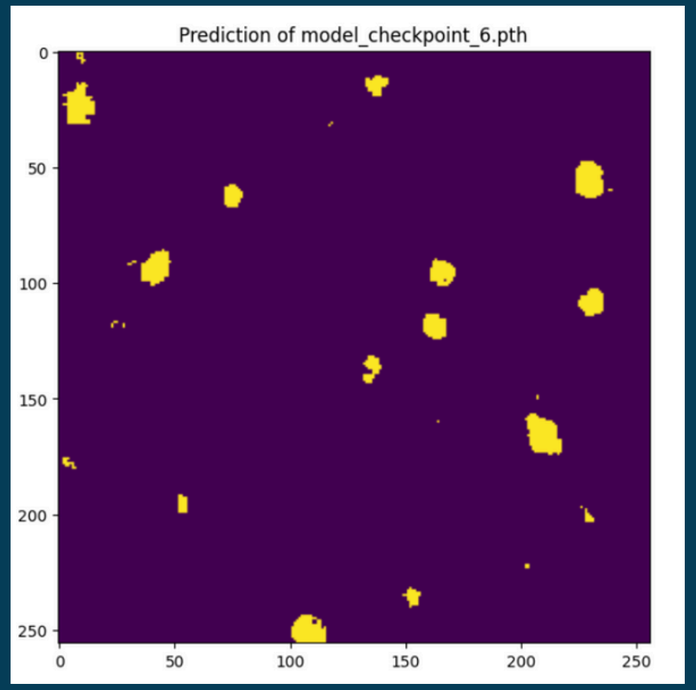


Image
0
50
100
150
200
250
0
50
100
150
200
250


Prediction of model_checkpoint_1.pth
0
50
100
150
200
250
0
50
100
150
200
250


Prediction of model_checkpoint_14.pth
0
50
100
150
200
250
0
50
100
150
200
250


true mask
0
50
100
150
200
250
0
50
100
150
200
250


Prediction of model_checkpoint_6.pth
0
50
100
150
200
250
0
50
100
150
200
250

MERCI !